

Operations, Environment & Readiness

The practical companion to `architecture.md`: every environment variable, how to run the tests, the dev sign-in accounts, what is not built yet, and the checklist to reach production. Email specifics live in `email-ses.md`.

1 Environment variables

Parsed once at boot by `src/lib/env.ts` (Zod) and cached. It is **fail-closed**: the app refuses to start if a required value is missing or malformed, rather than booting into a broken state. `.env.example` ships every unsafe value in its **failing** state — a verbatim copy will not boot.

VARIABLE	REQUIRED	DEFAULT	NOTES
DATABASE_URL	✅ always	—	Postgres connection string
TEST_DATABASE_URL	int tests only	—	Used by pnpm test:int
SESSION_SECRET	✅ always	—	JWT signing key, ≥ 32 chars . 🔒 secret
AUTH_MODE	✅ always	— (no default)	stub cgsauth — an explicit choice
CGS_AUTH_BASE_URL	—	https://teamcgs.io	UAT: https://uat.teamcgs.io
CGS_AUTH_API_KEY	when cgsauth	—	CGS x-api-key. 🔒 secret, server-only
ROSTER_ALLOWLIST	prod	—	Comma-separated Domain IDs / emails allowed to sign in
ROSTER_ALLOW_ALL_WHEN_EMPTY	—	false	Only "true" admits everyone when the roster is empty. Dev only
APP_URL	—	http://localhost:3000	Public origin
NODE_ENV	—	development	development test production
EMAIL_MODE	—	stub	stub (no-send) ses. stub refuses to run in prod
SES_REGION	when ses	—	e.g. ap-southeast-1
SES_ACCESS_KEY_ID	when ses	—	🔒 secret
SES_SECRET_ACCESS_KEY	when ses	—	🔒 secret
EMAIL_FROM	when ses	—	A verified SES identity, e.g. Van Reservation <no-reply@...>
EMAIL_REPLY_TO	—	—	Optional Reply-To

🔒 = keep out of the repo; set only in an untracked .env or a secrets manager.

Fail-closed rules to know:

- AUTH_MODE=cgsauth requires CGS_AUTH_API_KEY, or the app won't boot.
- EMAIL_MODE=ses requires SES_REGION, SES_ACCESS_KEY_ID, SES_SECRET_ACCESS_KEY, EMAIL_FROM, or the app won't boot.
- An **empty** ROSTER_ALLOWLIST **denies everyone** unless ROSTER_ALLOW_ALL_WHEN_EMPTY=true — this is intentional (FR-2).

Minimal local .env (stub, no external services)

```
DATABASE_URL=postgres://van:van@localhost:5433/van_reservation
SESSION_SECRET=$(openssl rand -base64 48) # any ≥32-char string
AUTH_MODE=stub
ROSTER_ALLOW_ALL_WHEN_EMPTY=true # dev only – lets the fixtures sign in
# EMAIL_MODE defaults to stub – no SES needed locally
```

2 Testing

Two suites, run separately (see `vitest.config.ts` / `vitest.int.config.ts`).

	PNPM TEST (UNIT)	PNPM TEST:INT (INTEGRATION)
Files	36 (*.test.ts(x))	6 (*.int.test.ts)
Needs Postgres?	No	Yes
Covers	pure rules, providers/transport (faked), components, env	DB client, migrations, schema CHECKS, whitelist seed, live authenticate()

- Unit tests need nothing but `pnpm install` — no DB, no network (SES/CGS are faked). Fast.
- Integration tests connect to Postgres (`pnpm db:up`) and **auto-create** `van_reservation_test` on first run, then migrate it.
- Other gates: `pnpm typecheck` (`tsc --noEmit`), `pnpm lint` (Biome), `pnpm build`.

3 Test accounts (stub mode only)

These are the fixtures in `src/modules/auth/dev-identities.ts`, resolved **only when** `AUTH_MODE=stub`. The stub **ignores the password** — the Domain ID alone selects the identity (type anything into the password field).

Admins (role `admin_support` → `/dashboard`):

DOMAIN ID	NAME	SITE
IB10001	Ivy Balandra	Iloilo
JG10002	Jezreel Mariz Gromia	Iloilo
AM10003	Angel Grace Mateo	Iloilo
ZB10004	Zarra Crist Bartolo	Manila
ND10005	Norlen Denonong	Manila
SL10006	Shaw Lim	Manila
RE10007	Ruwi Joy Eribal	All

Associates (role associate → /):

DOMAIN ID	NAME
AB12345	Juan Dela Cruz
CD67890	Maria Santos

To use them, two gates must be satisfied:

1. **Sign-in gate (roster):** every sign-in is checked against ROSTER_ALLOWLIST. With an empty roster, set ROSTER_ALLOW_ALL_WHEN_EMPTY=true locally or **nobody** — admin or associate — can log in.
2. **Admin role gate (whitelist):** the admin role comes from the admin_whitelist table, not the fixture. To get an admin in dev you must have run `pnpm db:migrate && pnpm db:seed`; without the seed the table is empty and everyone resolves to associate.

In cgsauth mode these fixtures do nothing — you need a real CGS Domain ID + password.

4 What's not built yet

Scaffolding and rules exist ahead of the flows that use them; know the gaps:

- **Reservations are fixture-backed.** modules/reservations / modules/drivers hold the rules; the admin/requestor UIs render against fixtures. There are no reservation **read/write** endpoints wired to Postgres yet.

- **Email is scaffolded but unwired.** `sendBookingConfirmation()` is built and tested but no event calls it — the booking write-flows that would (Approved / Rejected / ...) don't exist yet.
 - **Roles are cached in the session JWT** and not re-checked per request, so revoking `admin_support` lags up to the 8-hour token TTL.
 - **AUTH_MODE=stub and dev-identities.ts** are development-only; production uses `cgsauth` and this fixture file should be deleted once stub mode retires.
 - **admin_whitelist seed uses placeholder @example.invalid emails** (`modules/auth/admin-whitelist-seed.ts`).
 - **Elevance Sans** is not licensed (falls back to Inter); the van hero photo's licensing is unconfirmed.
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5 Production readiness checklist

SECRETS & CONFIG

- ☐ `SESSION_SECRET` set to a fresh ≥32-char random value (not the example).
- ☐ `NODE_ENV=production`, `APP_URL` set to the real origin.
- ☐ All  secrets injected via the platform's secret store, never the repo.

AUTH

- ☐ `AUTH_MODE=cgsauth` with the real `CGS_AUTH_API_KEY` (and correct `CGS_AUTH_BASE_URL`).
- ☐ `ROSTER_ALLOWLIST` populated; confirm `ROSTER_ALLOW_ALL_WHEN_EMPTY` is unset/false (an empty roster must deny everyone).
- ☐ Replace the placeholder emails/Domain IDs in `admin-whitelist-seed.ts` with real admins, then `pnpm db:seed`.
- ☐ Decide on immediate role-revocation (session table or short TTL) before admin write operations ship.

EMAIL

- ☐ `EMAIL_MODE=ses + SES_REGION / credentials / EMAIL_FROM`.
- ☐ Verify the SES sender identity (+ DKIM) and leave the SES sandbox.
- ☐ Wire `sendBookingConfirmation()` (and any other notices) to real events.

DATABASE

- ☐ Managed Postgres provisioned; DATABASE_URL points at it.
- ☐ `pnpm db:migrate` run against it; backups configured.

PLATFORM

- ☐ License + install the Elevance Sans web font; confirm van photo licensing.
- ☐ Plan the `middleware.ts` → `proxy.ts` migration (Next 16 deprecation) with an import-restriction lint rule (see `architecture.md` §13).
- ☐ `pnpm typecheck && pnpm lint && pnpm test && pnpm test:int && pnpm build` all green in CI.